

STAT 252 — In-Class Experiments

Guided Analysis Workbook

Cal Poly San Luis Obispo

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How to Use This Workbook

This workbook walks through three in-class experiments using the **SSVA framework** (Summary statistics → Visualization → Analysis). For each experiment we will:

1. **State the research question.**
2. **Look at the raw data** directly in the Google Sheet.
3. **Examine summary statistics.**
4. **Interpret the visualization.**
5. **Run the analysis** (hypothesis test + confidence interval).
6. **Answer SSVA guiding questions** as a class.
7. **Answer independent conceptual questions** on your own.

Session code: sec-10

Class Google Sheet: [open in new tab](#)

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Tip

Before each experiment: open the class Google Sheet linked above and click the tab for that experiment (reaction, memory, or framing) to see the raw data you helped collect.

1. Experiment 1 — Two Independent Population Means

1.1 Research Question

Does listening to music while performing a simple task change how quickly people can react?

1.2 Step 1 — Look at the Raw Data

Note

Open the class Google Sheet and click the reaction tab. Look at how each row is a single student, with their assigned group and their average reaction time across 5 trials. How many rows do you see? How are students split between the two groups?

First few rows of the reaction time data (for reference)

Group	Average RT (ms)
music	429
music	453
music	576
silence	429
silence	453
silence	576

1.3 Step 2 — Summary Statistics

Summary statistics by group

Group	n	Mean (ms)	SD (ms)	Median (ms)	Min	Max
music						
silence						

1.4 Step 3 — Visualization

Describe Visualization:

1.5 Step 4 — Analysis

Welch two-sample t-test: music vs. silence

Quantity	Value
Test statistic (t)	
Degrees of freedom	
p-value	
95% CI lower	
95% CI upper	

1.6 SSVA Guiding Questions — Experiment 1

Q1. How many variables are present in this experiment?

Your answer:

Q2. What are the types of each variable?

Your answer:

Q3. What are the response and explanatory variables?

Your answer:

Q4. What relationship is shown in the visualization? Which group appears to react faster on average, and how much overlap is there between the two distributions?

Your answer:

Q5. What do the measures of center and variability tell you? Is one group more variable than the other?

Your answer:

Q6. Based on the visualization and summary statistics alone, what is your *initial* conclusion about whether music affects reaction time? (Don't look at the p-value yet.)

Your answer:

1.7 Independent Conceptual Questions — Experiment 1

C1. What is the statistical question?

C2. What is the population of interest?

C3. What is the sample?

C4. What is the observational unit?

C5. Describe all variables in the study (name, type, role).

C6. What potential confounding variables exist?

C7. Is this an experiment or an observational study? How do you know?

C8. What design improvements could enhance the study?

C9. State the null and alternative hypotheses (in both words and symbols).

C10. Based on the test statistic and p-value, what do you conclude about H_0 ?

C11. If the confidence interval were not provided, could you infer whether 0 would be in it based on the hypothesis test decision? Explain.

2. Experiment 2 — Paired t-Test

2.1 Research Question

Does learning a simple chunking strategy improve short-term recall of a 14-digit sequence?

2.2 Step 1 — Look at the Raw Data

Note

Open the class Google Sheet and click the memory tab. Each row is a single student with two scores: score_before (Round 1, no chunking) and score_after (Round 2, with chunking). Look at a few rows — does the “after” score tend to be higher, lower, or the same as “before” for each student?

First few rows of the memory data (for reference)

Before (Round 1)	After (Round 2)	Difference
6	6	0
7	5	-2
10	6	-4
20	15	-5

2.3 Step 2 — Summary Statistics

Summary statistics

Quantity	Value
n (students)	
Mean score, Round 1 (before)	
Mean score, Round 2 (after)	
Mean within-student difference (after – before)	
SD of differences	
Min difference	
Max difference	

2.4 Step 3 — Visualization

Describe Visualization:

2.5 Step 4 — Analysis

Paired t-test: after – before chunking

Quantity	Value
Test statistic (t)	
Degrees of freedom	
p-value	
95% CI lower	
95% CI upper	

2.6 SSVA Guiding Questions — Experiment 2

Q1. How many variables are there in this experiment, and why is this design different from Experiment 1?

Your answer:

Q2. What are the types of the variables, and what is the key *derived* variable for the paired analysis?

Your answer:

Q3. What are the explanatory and response variables?

Your answer:

Q4. What does the visualization show? What does a line that slopes upward mean for an individual student? What about a flat or downward line?

Your answer:

Q5. What does the mean difference tell you? What does the standard deviation of the differences tell you?

Your answer:

Q6. Based on SSVA alone, what is your initial conclusion about whether chunking helps?

Your answer:

2.7 Independent Conceptual Questions — Experiment 2

C1. What is the statistical question?

C2. What is the population of interest?

C3. What is the sample?

C4. What is the observational unit?

C5. Describe all variables in the study.

C6. What potential confounding variables exist? Consider what's special about a before/after design.

C7. Is this an experiment or an observational study?

C8. What design improvements could enhance the study?

C9. State the null and alternative hypotheses.

C10. Based on the test statistic and p-value, what do you conclude about H_0 ?

C11. If the confidence interval were not provided, could you infer whether 0 is in it from the test decision?

3. Experiment 3 — Two Population Proportions

3.1 Research Question

Does the way a medical outcome is framed — in terms of survival versus mortality — change whether people recommend the treatment?

3.2 Step 1 — Look at the Raw Data

Note

Open the class Google Sheet and click the framing tab. Each row is a single student: the group column tells you which framing they saw (positive = 90% survive, negative = 10% die), and the choice column records their yes/no recommendation. Roughly how many students are in each group? How many said “yes” in each?

First few rows of the framing data (for reference)

Frame	Choice
positive	yes
negative	no
positive	yes
negative	no

3.3 Step 2 — Summary Statistics

Recommendation rates by framing condition

Frame	n	# Yes	# No	Proportion ‘yes’
positive				
negative				

3.4 Step 3 — Visualization

Describe Visualization:

3.5 Step 4 — Analysis

Two-proportion z-test: positive vs. negative frame

Quantity	Value
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Test statistic (z)	
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p-value	
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95% CI lower	
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95% CI upper	
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3.6 SSVA Guiding Questions — Experiment 3

Q1. How many variables are there, and what type is each?

Your answer:

Q2. What are the response and explanatory variables?

Your answer:

Q3. What does the bar chart show? Which frame produced a higher recommendation rate?

Your answer:

Q4. For categorical data like this, we don't compute a mean and SD. What summary statistics *do* we use, and why?

Your answer:

Q5. Based on SSVA alone, what is your initial conclusion about whether framing affects recommendations?

Your answer:

3.7 Independent Conceptual Questions — Experiment 3

C1. What is the statistical question?

C2. What is the population of interest?

C3. What is the sample?

C4. What is the observational unit?

C5. Describe all variables in the study.

C6. What potential confounding variables exist?

C7. Is this an experiment or an observational study?

C8. What design improvements could enhance the study?

C9. State the null and alternative hypotheses.

C10. Based on the test statistic and p-value, what do you conclude about H_0 ?

C11. If the confidence interval were not given, could you infer whether 0 is inside it from the test decision?
